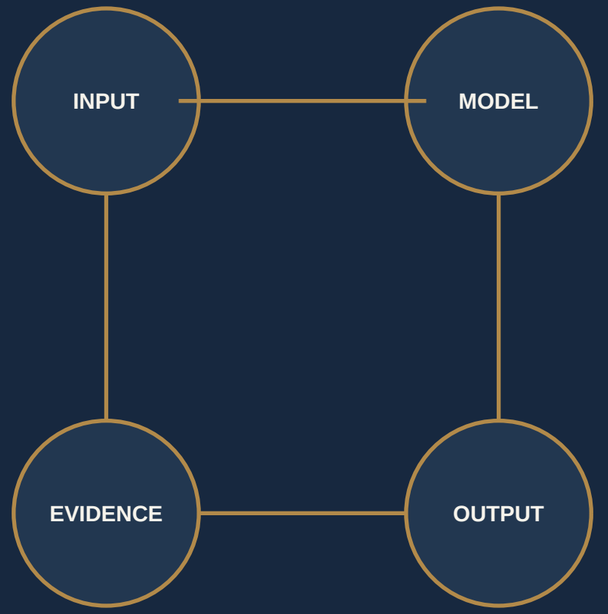
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Cognitive Robotics

AI.305 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define perception.
- Apply key methods including reasoning.
- Build or analyze a practical system involving memory.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Perception

02

METHODS

Reasoning

03

SYSTEMS

Memory

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Perception

Perception

02 · Reasoning

Reasoning

03 · Memory

Memory

04 · Learning

Learning

05 · Planning

Planning

06 · Decision-making and cognitive

Decision-making and cognitive architectures for robots

07 · Mandatory practical: Develop

Mandatory practical: Develop a cognitive robotic agent capable of perception

08 · Reasoning and action

Reasoning and action

Core ideas, made explicit

AI.305 • Distinguish definitions, assumptions and evidence

- Perception
- Reasoning
- Memory
- Learning
- Planning

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

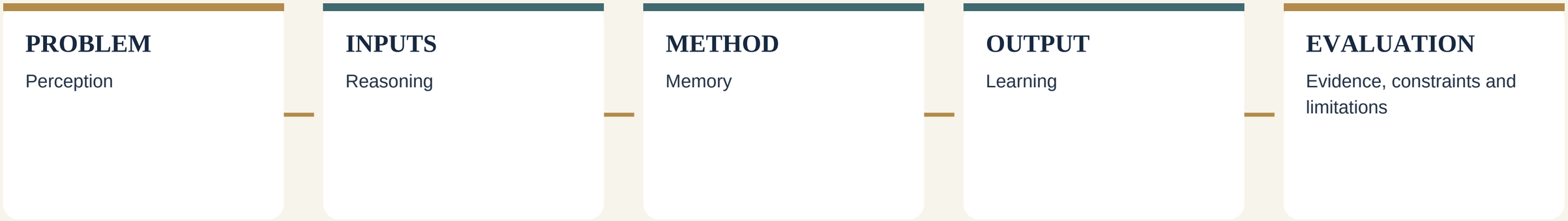
AI.305 • Distinguish definitions, assumptions and evidence

- Decision-making and cognitive architectures for robots
- Mandatory practical: Develop a cognitive robotic agent capable of perception
- Reasoning and action

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.



Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving perception.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using perception.
- Implement or simulate a baseline using reasoning.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.